

Christopher P. Hurley

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Summary

Expert software engineer specializing in system-level development for SoC, storage, memory, and firmware. Proven strengths in CI/CD integration, cross-organizational collaboration, and pragmatic process improvement across modeling, validation, and hardware bring-up. Adept at navigating complex team interfaces to de-risk schedules and accelerate developer velocity.

Skills

- Languages: C++, C, Python, SystemVerilog, Verilog, JavaScript, Ruby
- Hardware/Modeling: SystemC (TLM-2.0), UVM, RTL/DPI integration
- Storage/Embedded: SSD/NAND media, firmware, embedded systems
- Tools: CMake, GoogleTest, CI/CD, Git, clang-format, clang-tidy
- Platforms: Linux, Windows, macOS; virtualization and containers
- Practices: Agile/Scrum, clean code, test-driven development, documentation

Experience

Qualcomm — Software Engineer (2025-04 – Present; Remote)

- Led bring-up of a full SoC virtual platform and integrated it into CI, creating a documented workflow that accelerates new platform additions.
- Built a SystemC unit test framework (GoogleTest + CMake) with parallel CI execution and quality gates.
- Standardized code quality with clang-format, clang-tidy, and stricter compiler warnings, enforced via CI.
- Implemented platform-aware register mapping in early boot flows (e.g., TF-A to UEFI), stabilizing simulation boot.
- Designed a scalable, router-based dynamic address decoder (TLM-2.0), reducing lookup complexity for large address maps.
- Automated Markdown-to-wiki publishing in CI to ensure virtualization and platform documentation remains current.
- Contributed upstream improvements to SystemC component libraries for modern C++ (Clang/LLVM 17, C++20).

Solidigm — Software Engineer (2021-07 – 2025-03; Remote)

- Developed NAND SSD firmware and media models in C/C++ with Python tooling.
- Aligned and extended NAND media models to interoperate with partner stacks, shaping a unified modeling strategy.
- Designed and deployed simulation-driven CI workflows, expanding test coverage and enabling early regression detection.
- Integrated SoC and NAND models for shift-left development, enabling an ahead-of-schedule functional demo on a new SoC/NAND combination.
- Implemented and optimized a QLC-focused garbage-collection algorithm with DRAM caching, improving I/O stability and performance.
- Modernized simulation toolchains with compiler updates and reproducible builds to streamline developer workflows.
- Established initial power-on firmware and bring-up infrastructure across multiple programs and form factors.
- Facilitated cross-team alignment and clarified interfaces to reduce friction between modeling, firmware, and validation stakeholders.

Intel — Software Engineer (2017-03 – 2021-06; Folsom, CA)

- Built C++ models for 3D XPoint and NAND with TLM-2.0 and SystemC/SystemVerilog interfaces to support architecture, RTL, and firmware development.

- Designed a unified NAND model core leveraged across RTL validation, emulation, and virtual platforms, integrating it with SSD VP and QoS models.
- Collaborated with architecture and design teams to model new NAND features, enabling earlier firmware development and pre-silicon feedback.
- Enhanced 3D XPoint timing workflows by replacing manual spreadsheets with version-controlled Python tooling, enabling rapid what-if analysis.
- Architected a high-performance, multithreaded C++/SV-DPI testbench for transactor validation.
- Led team planning, Scrum processes, and DevOps initiatives; mentored new engineers and performed code reviews to improve quality and velocity.

Oracle — Cloud Application Developer (2016-07 – 2017-03; Rocklin, CA)

- Migrated an internal sales configuration system to Oracle CPQ Cloud.
- Developed front-end (JavaScript/HTML), back-end (BML/SQL), and automation/testing scripts in Python.
- Led a licensing product selection tool from requirements gathering through UAT and go-live.
- Implemented Jira for project management and authored process documentation and functional specifications.

Intel — Software Engineer (2014-04 – 2016-07; Folsom, CA)

- Architected a SystemC/Python component testbench that enabled high-level testing via customer-facing interfaces, reducing test times from minutes to seconds.
- Integrated C++ unit testing (GoogleTest), reducing targeted test execution from seconds to milliseconds.
- Consolidated builds into a single cross-platform CMake configuration with automated testing and packaging (CTest/CPack), cutting build times from ~5 minutes to ~30 seconds.
- Unified multiple disparate models into one configurable core with role-specific interfaces, coordinating parameters with design and architecture.
- Drove adoption of Scrum with regular release cycles; facilitated daily stand-ups, sprint planning, and customer syncs.

Intel — System Test Engineer (2009-09 – 2014-03; Folsom, CA)

- Developed C++ manufacturing test flows for SAS, Fibre Channel, and PCIe SSDs from initial development through high-volume manufacturing.
- Migrated test infrastructure from a limited Windows PC setup to a custom, highly parallel Linux-based system.
- Led joint development with tester vendors on new products and drive interfaces, increasing code reuse and collaboration across product lines.
- Established Jira for issue management and Confluence for team collaboration.
- Partnered with firmware teams to debug manufacturing issues and implement features to reduce test costs.

Earlier Experience

- Intel — Design Engineering Intern (2008-06 – 2009-08): Assisted with RTL design, synthesis, verification, and timing. Developed and taught a SystemVerilog curriculum for engineers.
- UC Davis — Micromouse Robotics Team Lead (2008-09 – 2009-06): Led a 5-person team to design, build, and program an autonomous maze-solving robot, winning the 2009 UC Davis Micromouse competition.

Education

B.S., Computer Engineering — University of California, Davis (2009)